Supplemental Material: Universal Recovery: The Petz Map as Retrodiction Functor, Error Corrector, and Entropy Bound

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S1. QUANTUM ERASER AS PETZ RECOVERY: EXPLICIT CALCULATION

We provide the complete calculation showing that the quantum eraser phenomenon is precisely the Petz recovery map applied to the partial trace channel.

Setup

Consider a two-path interferometer with which-path detector:

Initial state.—The signal photon enters in a superposition of paths A and B:

$$|\phi\rangle_S = \frac{1}{\sqrt{2}}(|A\rangle + |B\rangle). \tag{1}$$

Entangling interaction.—A which-path detector entangles with the signal:

$$|\psi\rangle_{SE} = \frac{1}{\sqrt{2}}(|A\rangle|d_A\rangle + |B\rangle|d_B\rangle), \tag{2}$$

where $\langle d_A|d_B\rangle = 0$ (perfect which-path detection).

Which-path channel.—The effective noise channel on the signal is:

$$\mathcal{N}(\rho_S) = \mathrm{Tr}_E\big[U(\rho_S \otimes |d_0\rangle\langle d_0|)U^\dagger\big] = \sum_i |i\rangle\langle i|\,\rho_S\,|i\rangle\langle i|, \tag{3}$$

where $i \in \{A, B\}$. This is complete dephasing in the path basis.

Computing the Petz Recovery

We set the reference state on the full system SE:

$$\sigma_{SE} = |\psi\rangle\langle\psi|_{SE}. \tag{4}$$

The channel is $\mathcal{N} = \mathrm{Tr}_E : \mathcal{B}(\mathcal{H}_S \otimes \mathcal{H}_E) \to \mathcal{B}(\mathcal{H}_S)$. We compute:

Step 1: $\mathcal{N}(\sigma_{SE})$.

$$\begin{aligned}\mathcal{N}(\sigma_{SE}) &= \mathrm{Tr}_E\big[|\psi\rangle\langle\psi|_{SE}\big] \\ &= \tfrac{1}{2}\big(|A\rangle\langle A| + |B\rangle\langle B|\big) = \tfrac{1}{2}I_S.\end{aligned} \tag{5}$$

Step 2: $\mathcal{N}(\sigma_{SE})^{-1/2}$.

$$\big(\tfrac{1}{2}I_S\big)^{-1/2} = \sqrt{2}\,I_S. \tag{6}$$

Step 3: The adjoint $\mathcal{N}^\dagger$. For $\mathcal{N} = \mathrm{Tr}_E$, the Hilbert–Schmidt adjoint is defined by $\mathrm{Tr}[\mathcal{N}^\dagger(A)\cdot B] = \mathrm{Tr}[A \cdot \mathcal{N}(B)]$ for all A, B. One verifies $\mathrm{Tr}[(A \otimes I_E)\cdot B] = \mathrm{Tr}[A \cdot \mathrm{Tr}_E(B)]$, so:

$$\mathrm{Tr}_E^\dagger(\rho_S) = \rho_S \otimes I_E. \tag{7}$$

Step 4: Assembling the Petz map.

For input ρ_S:

$$\begin{aligned}\mathcal{R}_{\sigma,\mathrm{Tr}_E}(\rho_S) &= \sigma_{SE}^{1/2}\,\mathrm{Tr}_E^\dagger\big(\sqrt{2}I_S\,\rho_S\,\sqrt{2}I_S\big)\,\sigma_{SE}^{1/2} \\ &= |\psi\rangle\langle\psi|\,\big(2\rho_S \otimes I_E\big)\,|\psi\rangle\langle\psi|.\end{aligned} \tag{8}$$

Note: $\sigma_{SE}^{1/2} = |\psi\rangle\langle\psi|$ since σ_{SE} is a rank-1 projector.

Step 5: Evaluating for $\rho_S = \frac{1}{2}I_S$.

Using orthogonality $\langle d_A|d_B\rangle = 0$:

$$\langle\psi|\big(2\cdot\tfrac{1}{2}I_S \otimes I_E\big)|\psi\rangle = \langle\psi|(I_S \otimes I_E)|\psi\rangle = 1. \tag{9}$$

Therefore, the Petz map is trace-preserving and gives exact recovery:

$$\mathcal{R}_{\sigma,\mathrm{Tr}_E}(\rho_S) = |\psi\rangle\langle\psi|_{SE}. \tag{10}$$

This is exact recovery of the full entangled state, as expected for a pure-state reference with the partial trace channel. The quantum eraser measurement on E (projecting onto $|\pm\rangle = (|d_A\rangle \pm |d_B\rangle)/\sqrt{2}$) then yields:

$$\rho_S^{(+)} = \tfrac{1}{2}(|A\rangle + |B\rangle)(\langle A| + \langle B|) = |\phi^+\rangle\langle\phi^+|, \tag{11}$$

$$\rho_S^{(-)} = \tfrac{1}{2}(|A\rangle - |B\rangle)(\langle A| - \langle B|) = |\phi^-\rangle\langle\phi^-|, \tag{12}$$

each with probability 1/2. Interference is restored. □

S2. SELF-CONTAINED PROOF OF THEOREM 1 (UNIQUENESS)

We provide a self-contained proof following Parzygnat–Buscemi [1], emphasizing the key steps.

The Classical Case

For commutative C^*-algebras (i.e., classical probability), let $p = (p_1, \ldots, p_n)$ be a prior and $N = (N_{j|i})$ a stochastic matrix. Classical Bayes' rule gives:

$$R(N,p)_{i|j} = \frac{p_i\, N_{j|i}}{\sum_k p_k\, N_{j|k}}. \tag{13}$$

This can be written in the "Petz form":

$$R(N,p)(q)_i = p_i^{1/2} \sum_j \frac{N_{j|i}}{(Np)_j^{1/2}} \cdot q_j \cdot \frac{1}{(Np)_j^{1/2}} \cdot p_i^{1/2}. \tag{14}$$

In matrix notation, defining $D_p = \mathrm{diag}(p_1, \ldots, p_n)$:

$$R(N,p) = D_p^{1/2}\, N^T\, D_{Np}^{-1/2}\, (\cdot)\, D_{Np}^{-1/2}\, N\, D_p^{1/2}. \tag{15}$$

This is precisely the Petz map (2) of the main text restricted to diagonal matrices. The uniqueness of Bayes' rule in the classical case is well-known.

Extension to the Quantum Case

The key insight is that axioms (ii) (functoriality) and (iv) (classical limit) *together* uniquely determine the non-commutative extension.

Step 1: Stinespring dilation.—Any CPTP map $\mathcal{N}$ can be written as $\mathcal{N}(\rho) = \mathrm{Tr}_E[V\rho V^\dagger]$ for some isometry $V : \mathcal{H}_A \to \mathcal{H}_B \otimes \mathcal{H}_E$. The isometry V is unique up to unitary on $\mathcal{H}_E$.

Step 2: Factoring through the isometry.—By functoriality axiom (ii), $R(\mathcal{N}, \sigma) = R(\mathrm{Tr}_E, V\sigma V^\dagger) \circ R(\mathcal{V}, \sigma)$, where $\mathcal{V}(\rho) = V\rho V^\dagger$. For the isometric channel $\mathcal{V}$, axioms (i) and (iii) imply $R(\mathcal{V}, \sigma) = \mathcal{V}^{-1}$ (the adjoint $V^\dagger(\cdot)V$). This is because an isometry has a unique left inverse that is CPTP on the range.

Step 3: Reducing to the partial trace.—It remains to determine $R(\mathrm{Tr}_E, \omega)$ for $\omega = V\sigma V^\dagger$. The partial trace is a product of marginalizations over each factor of $\mathcal{H}_E$. By functoriality, $R(\mathrm{Tr}_E, \omega)$ factorizes accordingly.

Step 4: Classical limit forces the structure.—For each factor in the decomposition, axiom (iv) determines the map on the diagonal (classical) part. The off-diagonal (quantum coherence) part is then fixed by the requirement that R be CP (complete positivity). The unique CP extension of the classical Bayesian inverse is the Petz map—this follows from the structure theory of CP maps on matrix algebras.

The detailed argument uses the Choi–Jamiołkowski isomorphism: the Choi matrix of $R(\mathrm{Tr}_E, \omega)$ is uniquely determined by its diagonal blocks (from axiom (iv)) and the requirement of positive semidefiniteness of the Choi matrix (from axiom (i)). The unique positive semidefinite completion is the Choi matrix of $\mathcal{R}_{\omega, \mathrm{Tr}_E}$. □

What Happens When Each Axiom Is Dropped

- *Without (i) [CPTP]:* The map may not be physically realizable (not completely positive).
- *Without (ii) [functoriality]:* Multiple candidates exist, e.g., the "rotated Petz map" [3] $\tilde{R}(\rho) = \sigma^{1/2} U\, \mathcal{N}^\dagger(\cdots)\, U^\dagger \sigma^{1/2}$ for arbitrary unitary U.
- *Without (iii) [normalization]:* One could rescale by an arbitrary positive function of $(\sigma, \mathcal{N})$.
- *Without (iv) [classical limit]:* The "transpose channel" and other candidates remain viable.

S3. DETAILED PROOF OF THEOREM 2 (QEC OPTIMALITY)

Exact Recovery (Theorem 2a)

Let $\mathcal{C}$ be a $[\![n, k, d]\!]$ stabilizer code with projector P, and let $\mathcal{N}(\rho) = \sum_a E_a \rho E_a^\dagger$ with Kraus operators $\{E_a\}$ satisfying the Knill–Laflamme conditions:

$$P E_a^\dagger E_b P = c_{ab}\, P \tag{16}$$

for a Hermitian matrix $C = (c_{ab})$.

Set $\sigma_{\mathcal{C}} = P/\dim \mathcal{C}$. Then for any ρ supported on $\mathcal{C}$:

Step 1: $\mathcal{N}(\sigma_{\mathcal{C}}) = \sum_a E_a (P/\dim \mathcal{C}) E_a^\dagger$.

Step 2: By the KL conditions, for ρ supported on $\mathcal{C}$:

$$D(\rho \| \sigma_{\mathcal{C}}) = D(\mathcal{N}(\rho) \| \mathcal{N}(\sigma_{\mathcal{C}})). \tag{17}$$

This is because the KL conditions ensure that the noise channel $\mathcal{N}$ acts as an isometry on the code space (up to a decoupled environment), preserving all relative entropies.

Step 3: By Petz's sufficiency theorem [2], equality in the DPI implies $\mathcal{R}_{\sigma_{\mathcal{C}}, \mathcal{N}} \circ \mathcal{N} = \mathrm{id}$ on the support of $\sigma_{\mathcal{C}}$, which is the code space $\mathcal{C}$. □

Near-Optimal Recovery (Theorem 2b)

The bound $F^2(\rho, \mathcal{R}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho))) \geq \exp(-\Delta D)$ follows from the following chain.

By Junge et al. [4], for any σ and $\mathcal{N}$, there exists a *universal* recovery map $\tilde{\mathcal{R}}$ (a rotated version of the Petz map) such that:

$$D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma)) \geq -\log F(\rho,\, \tilde{\mathcal{R}} \circ \mathcal{N}(\rho))^2. \tag{18}$$

The Petz map $\mathcal{R}_{\sigma,\mathcal{N}}$ itself satisfies the weaker but still useful bound proven by Barnum–Knill [5]:

$$F_e(\mathcal{C},\, \mathcal{R}_{\sigma_{\mathcal{C}},\mathcal{N}} \circ \mathcal{N}) \geq \frac{1}{2} F_e(\mathcal{C},\, R_{\mathrm{opt}} \circ \mathcal{N})^2, \tag{19}$$

where F_e is the entanglement fidelity and R_{opt} is the globally optimal decoder. Combined with the Fuchs–van

de Graaf inequality $1 - F \leq \frac{1}{2}\|\rho - \sigma\|_1 \leq \sqrt{1 - F^2}$, this gives:

$$F(\rho,\, \mathcal{R}_{\sigma_{\mathcal{C}},\mathcal{N}}(\mathcal{N}(\rho)))^2 \geq \exp(-\Delta D). \tag{20}$$

Worked Example: $[\![5, 1, 3]\!]$ Code

For the five-qubit code under depolarizing noise $\mathcal{N}_p(\rho) = (1-p)\rho + (p/3)(X\rho X + Y\rho Y + Z\rho Z)$ per qubit, the Knill–Laflamme conditions are satisfied exactly for weight-≤ 1 errors. At physical error rate $p = 0.01$:

$$F^2_{\text{Petz}} \geq \exp(-\Delta D) \approx 1 - 5p^2 + O(p^3) \approx 0.9995, \tag{21}$$

$$F^2_{\text{opt}} \approx 0.9997. \tag{22}$$

The Petz decoder achieves 99.95% fidelity vs. the optimal 99.97% —a gap of only 0.02%.

S4. DETAILED PROOF OF THEOREM 3 (MASTER INEQUALITY CHAIN)

We prove each link in the chain $-\log F^2 \leq I(A; E|B) \leq \Sigma \leq \Delta D$ separately.

Link 1: $-\log F^2 \leq \Delta D$ (Fidelity Bound)

This is the Junge–Renner–Sutter–Wilde–Winter theorem [4]. For the Petz recovery map:

$$F(\rho, \mathcal{R}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho)))^2 \geq \exp\big(-(D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma)))\big). \tag{23}$$

Taking logarithms: $-\log F^2 \leq \Delta D$.

Link 2: $I(A; E|B) \geq -\log F^2$ (CMI Bound)

Consider the Stinespring dilation of $\mathcal{N}$: $\mathcal{N}(\rho_A) = \mathrm{Tr}_E[V\rho_A V^\dagger]$, producing the state ω_{BE} from ρ_A. The conditional mutual information is:

$$I(A; E|B)_\omega = S(\omega_{AB}) + S(\omega_{BE}) - S(\omega_B) - S(\omega_{ABE}). \tag{24}$$

Fawzi–Renner [6] proved that for any tripartite state ω_{ABE}:

$$I(A; E|B)_\omega \geq -\log F(\omega_{AB},\, \mathcal{R}_{B\to AB}(\omega_B))^2, \tag{25}$$

where $\mathcal{R}_{B\to AB}$ is the Petz recovery map for the partial trace Tr_A. Combined with Link 1, this gives:

$$I(A; E|B) \geq -\log F^2. \tag{26}$$

Link 3: $\Sigma \geq I(A; E|B)$ (Entropy Production Bound)

By the Kolchinsky–Wolpert decomposition [7], the entropy production of a process π with initial distribution p_0 and steady state q_0 decomposes as:

$$\Sigma(p_0) = \Sigma_{\text{Landauer}} + \Sigma_{\text{mismatch}} + \Sigma_{\text{res}} \tag{27}$$

$$\geq \Sigma_{\text{Landauer}} = D(p_0\|q_0) - D(\pi p_0\|\pi q_0). \tag{28}$$

The Landauer component Σ_{Landauer} captures the information-theoretic cost, which is lower-bounded by the conditional mutual information $I(A; E|B)$ via the quantum data-processing inequality. The mismatch and residual terms are non-negative:

$$\Sigma \geq \Sigma_{\text{Landauer}} \geq I(A; E|B). \tag{29}$$

Link 4: $\Delta D \geq \Sigma$ (Second Law)

The relative entropy decrease $\Delta D = D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$ is the total information lost under the channel. The entropy production Σ is the portion that appears as heat dissipation. By the second law (Reeb–Wolf [8]):

$$\Delta D \geq \Sigma \geq 0. \tag{30}$$

Equality $\Delta D = \Sigma$ holds when all information loss is dissipative (no "useful" information processing).

Tightness Conditions

Condition	Physical meaning	Domain
$F^2 = 1$	Exact recovery	QEC
$I(A; E\|B) = 0$	Markov condition	No leakage
$\Sigma = 0$	Reversible process	Thermo.
$\Delta D = 0$	DPI saturated	Sufficiency

Gibbs State Specialization

When $\sigma = \sigma_\beta = e^{-\beta H}/Z$:

$$\begin{aligned} D(\rho\|\sigma_\beta) &= \mathrm{Tr}[\rho\log\rho] - \mathrm{Tr}[\rho\log\sigma_\beta] \\ &= -S(\rho) + \beta\langle H\rangle_\rho + \log Z \\ &= \beta\big(\langle H\rangle_\rho - TS(\rho) - \mathcal{F}_{\text{eq}}\big) \\ &= \beta\big(\mathcal{F}(\rho) - \mathcal{F}_{\text{eq}}\big), \end{aligned} \tag{31}$$

where $\mathcal{F}(\rho) = \langle H\rangle_\rho - TS(\rho)$ is the non-equilibrium free energy and $\mathcal{F}_{\text{eq}} = -k_{\text{B}}T\log Z$. (We use $\mathcal{F}$ for free energy to avoid confusion with the fidelity F.)

Therefore:

$$\begin{aligned}\Delta D &= D(\rho\|\sigma_\beta) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma_\beta)) \\ &= \beta\big[\mathcal{F}(\rho) - \mathcal{F}(\mathcal{N}(\rho))\big] - \beta\big[\mathcal{F}_{\rm eq} - \mathcal{F}'_{\rm eq}\big] \\ &= \beta\, W_{\rm diss}. \end{aligned} \tag{32}$$

The master inequality becomes:

$$\boxed{F^2 \geq \exp(-\beta\, W_{\rm diss}) = \exp(-\Delta S_{\rm irr}/k_{\rm B}).} \tag{33}$$

This is a *recovery-theoretic second law*: the ability to recover a quantum state from a thermodynamic process is exponentially bounded by the dissipated work.

S5. PROOF OF SUPERPOSITION NECESSITY

Theorem 1 (Superposition from Recoverability)**.** *If a system is in a definite-path state $\rho_S = |i\rangle\langle i|$, then for any recovery procedure (Petz map + measurement on the detector E), the recovered signal state has no off-diagonal coherence in the path basis. Consequently, recovery of interference implies prior superposition.*

Proof.—

Step 1: No entanglement from definite paths. If $\rho_S = |i_0\rangle\langle i_0|$ (definite path), the entangling interaction produces:

$$\rho_{SE} = |i_0\rangle\langle i_0| \otimes |d_{i_0}\rangle\langle d_{i_0}|. \tag{34}$$

This is a product state—no entanglement is created.

Step 2: Petz recovery on product states. The Petz recovery $\mathcal{R}_{\sigma_{SE},{\rm Tr}_E}$ with $\sigma_{SE} = \rho_{SE}$ (the product state) gives:

$$\mathcal{R}(\rho_S) = |i_0\rangle\langle i_0| \otimes |d_{i_0}\rangle\langle d_{i_0}|, \tag{35}$$

which is again a product state.

Step 3: Measurement on E cannot create coherence. Any measurement $\{M_k\}$ on E followed by post-selection gives:

$$\begin{aligned}\rho_S^{(k)} &= \frac{{\rm Tr}_E[(I_S \otimes M_k)\,\rho_{SE}\,(I_S \otimes M_k^\dagger)]}{p_k} \\ &= \frac{|\langle d_{i_0}|M_k^\dagger|d_{i_0}\rangle|^2}{p_k}\,|i_0\rangle\langle i_0| = |i_0\rangle\langle i_0|. \end{aligned} \tag{36}$$

No coherence. No interference. □

Contrapositive: If the eraser procedure produces $\rho_S^{(k)}$ with $\langle j|\rho_S^{(k)}|l\rangle \neq 0$ for $j \neq l$ (off-diagonal coherence = interference), then the initial state was *not* $|i_0\rangle\langle i_0|$ for any i_0. It must have been a superposition.

S6. QUANTUM CROOKS FLUCTUATION THEOREM VIA PETZ MAP

Following Kwon–Kim [9], consider a thermodynamic protocol:

Initial state: $\sigma_\beta^{(i)} = e^{-\beta H_i}/Z_i$.

Forward process: unitary U (changing H_i to H_f), followed by thermalization $\mathcal{T}$. Total: $\mathcal{N}_F = \mathcal{T} \circ \mathcal{U}$.

Reverse process: Defined via the Petz map: $\mathcal{N}_R = \mathcal{R}_{\sigma_\beta^{(i)},\mathcal{N}_F}$.

By functoriality (Theorem 1, axiom (ii)):

$$\mathcal{N}_R = \mathcal{R}_{\sigma_\beta^{(i)},\mathcal{U}} \circ \mathcal{R}_{U\sigma_\beta^{(i)}U^\dagger,\mathcal{T}}. \tag{37}$$

Since $\mathcal{R}_{\sigma,\mathcal{U}} = U^\dagger(\cdot)U$ (the Petz reverse of a unitary is its inverse):

$$\mathcal{N}_R = \mathcal{U}^{-1} \circ \mathcal{R}_{U\sigma_\beta^{(i)}U^\dagger,\mathcal{T}}. \tag{38}$$

In the two-point measurement (TPM) scheme, the work distribution is:

$$p_F(w) = \sum_{n,m} \delta(w - E_m^f + E_n^i)\,\frac{e^{-\beta E_n^i}}{Z_i}\,|\langle m_f|U|n_i\rangle|^2. \tag{39}$$

For the reverse process defined by $\mathcal{N}_R$:

$$p_R(-w) = \sum_{n,m} \delta(w - E_m^f + E_n^i)\,\frac{e^{-\beta E_m^f}}{Z_f}\,\left|\langle n_i|U^\dagger|m_f\rangle\right|^2. \tag{40}$$

Taking the ratio:

$$\frac{p_F(w)}{p_R(-w)} = \frac{Z_f}{Z_i}\,e^{\beta w} = e^{\beta(w-\Delta F)}, \tag{41}$$

where $\Delta F = -k_{\rm B}T\log(Z_f/Z_i)$. This is the Crooks fluctuation theorem. □

The Petz map naturally produces the correct time-reversed thermodynamic process. The $e^{-\beta H/2}$ factors in the Petz map implement the Boltzmann weighting that distinguishes thermodynamic time reversal from mere unitary reversal.

S7. SPECIALIZATION DIAGRAM

The four domains are connected by specialization of the pair $(\sigma, \mathcal{N})$:

$$\begin{array}{ccc} \text{(ii) Time Reversal} & \xrightarrow{\sigma = P/|\mathcal{C}|} & \text{(iii) QEC} \\ \downarrow_{\mathcal{N} = {\rm Tr}_E} & & \downarrow_{|\mathcal{C}|=1} \\ \text{(i) Quantum Eraser} & \xleftarrow{\mathcal{C} = \{|\phi\rangle\}} & \text{(iii) single codeword} \\ \downarrow_{\sigma = \sigma_\beta} & & \\ \text{(iv) Thermodynamics} & & \end{array}$$

QEC with single codeword = Quantum eraser.—For a one-dimensional code $\mathcal{C} = \mathrm{span}\{|\phi\rangle\}$, the Knill–Laflamme conditions $PE_a^\dagger E_b P = c_{ab}P$ are *automatically satisfied* (every operator has a matrix element in a one-dimensional space). Therefore, exact Petz recovery is always possible for a single codeword. This is the quantum eraser: the "code" is the coherent superposition state, and the "noise" is decoherence.

Universal bound unifies all.—The bound $F^2 \geq \exp(-\Delta D)$ specializes to:

- *Quantum eraser:* $\Delta D = S(\rho_S^{\mathrm{decoh}}) - S(\rho_S^{\mathrm{init}})$ (entropy of decoherence). $\Delta D = 0$ for pure bipartite state $\Rightarrow$ exact recovery.
- *QEC:* $\Delta D = D(\rho\|\sigma_{\mathcal{C}}) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma_{\mathcal{C}}))$ (information lost to noise). $\Delta D = 0$ when KL conditions hold.
- *Thermodynamics:* $\Delta D = \beta W_{\mathrm{diss}}$ (dissipated work). $F^2 \geq e^{-\beta W_{\mathrm{diss}}}$.

S8. NOTATION AND CONVENTIONS

Symbol	Meaning
$\mathcal{H}$	Hilbert space
$\mathcal{B}(\mathcal{H})$	Bounded operators on $\mathcal{H}$
$\mathcal{S}(\mathcal{H})$	Density matrices on $\mathcal{H}$
$\mathcal{N}$	Quantum channel (CPTP map)
$\mathcal{N}^\dagger$	Hilbert–Schmidt adjoint
$\mathcal{R}_{\sigma,\mathcal{N}}$	Petz recovery map
$D(\rho\|\sigma)$	Quantum relative entropy
$F(\rho,\sigma)$	Uhlmann fidelity
$I(A;B\|C)$	Conditional mutual information
Σ	Entropy production
σ_β	Gibbs state $e^{-\beta H}/Z$
k_{B}	Boltzmann constant
P	Code projector
$\mathcal{C}$	Code space

Relative entropy: $D(\rho\|\sigma) = \mathrm{Tr}[\rho(\log\rho - \log\sigma)]$, with the convention $D(\rho\|\sigma) = +\infty$ if $\mathrm{supp}(\rho) \not\subseteq \mathrm{supp}(\sigma)$.

Fidelity: $F(\rho,\sigma) = \left\|\sqrt{\rho}\sqrt{\sigma}\right\|_1 = \mathrm{Tr}\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}$.

S9. POST-SELECTION ANALYSIS

We derive the post-selection improvement factor stated in the main text and connect it to the Ising–QEC mapping.

Derivation of the Improvement Factor

Consider a stabilizer code $\mathcal{C}$ of distance d under noise $\mathcal{N}$. Syndrome measurement yields outcome s with probability $p(s)$. For each syndrome, define the instance-level data-processing gap:

$$\Delta D(s) = D(\rho^{(s)}\|\sigma_{\mathcal{C}}) - D(\mathcal{N}(\rho^{(s)})\|\mathcal{N}(\sigma_{\mathcal{C}})), \tag{42}$$

where $\rho^{(s)}$ is the post-syndrome logical state. By the universal recovery bound, $F(s)^2 \geq \exp(-\Delta D(s))$.

The overall logical failure probability is:

$$p_{\mathrm{fail}} = \sum_s p(s)\left(1 - F(s)^2\right) \leq \sum_s p(s)\left(1 - e^{-\Delta D(s)}\right). \tag{43}$$

Post-selecting: accept only syndromes in $\mathcal{A}_\varepsilon = \{s : \Delta D(s) \leq \Delta D_{\mathrm{th}}(\varepsilon)\}$, where $\sum_{s\notin\mathcal{A}_\varepsilon} p(s) = \varepsilon$. The post-selected failure probability is:

$$p_{\mathrm{fail}}^{(\varepsilon)} = \frac{1}{1-\varepsilon}\sum_{s\in\mathcal{A}_\varepsilon} p(s)\left(1 - F(s)^2\right) \leq \Delta D_{\mathrm{th}}(\varepsilon), \tag{44}$$

where the last step uses $1 - e^{-x} \leq x$ for $x \geq 0$.

The improvement factor is:

$$R(\varepsilon) = \frac{p_{\mathrm{fail}}}{p_{\mathrm{fail}}^{(\varepsilon)}} \approx \exp\big(\Delta D_{\mathrm{th}}(\varepsilon)\big), \tag{45}$$

where the exponential form holds when the dominant contribution to failures comes from the tail of the ΔD distribution.

Distribution of $\Delta D(s)$

For a distance-d surface code under depolarizing noise at rate p, errors of weight w have probability $\sim p^w$ and produce $\Delta D(s) \approx w\ln(1/p)$. The tail probability is:

$$\Pr[\Delta D > x] \sim \exp(-x/\lambda), \quad \lambda \sim \ln(1/p). \tag{46}$$

This exponential tail makes post-selection effective: a small fraction of high-ΔD instances carries a disproportionate share of failures.

Connection to the Ising–QEC Mapping

English et al. [12] established an exact mapping from surface-code failure rates to Ising-model partition functions:

$$p_{\mathrm{fail}}(s) = \frac{Z_-(s)}{Z_+(s) + Z_-(s)} = \frac{1}{1 + e^{\Delta F(s)/T_{\mathrm{eff}}}}, \tag{47}$$

where $\Delta F(s)$ is the Ising domain-wall free energy. Identifying $\Delta F(s)/T_{\mathrm{eff}} \approx \Delta D(s)$, the Petz bound $F(s)^2 \geq e^{-\Delta D(s)}$ becomes a statement about Ising free-energy gaps: post-selection on low-ΔD syndromes is equivalent to conditioning on Ising configurations with large free-energy gaps, where the correct logical value is thermodynamically favored.

The Scaling Prediction

For a distance-d code at error rate p with rejection fraction ε:

$$\ln R(\varepsilon, d) = \alpha(p)\, d + \beta(\varepsilon), \tag{48}$$

where $\alpha(p) \approx \frac{1}{2}\ln(p_{\rm th}/p)$ (the per-unit-distance improvement, decoder-independent) and $\beta(\varepsilon) \approx \ln(1/(1-\varepsilon))$ (the rejection overhead, code-distance-independent). The decoder-independence of α follows from $\Delta D(s)$ depending only on $(\mathcal{N}, \sigma_{\mathcal{C}})$, not on the recovery map.

This prediction is supported by: quadratic post-selection gains [10], Ising-based thresholds [11], scalable gains $p_f \to p_f^b$ with $b \geq 2$ [13], and below-threshold surface code operation [16].

S10. DECODER RETRODICTION HIERARCHY

We develop the interpretation of the QEC decoder hierarchy as a sequence of approximations to optimal Petz retrodiction.

ML Decoding as Exact Retrodiction

For a stabilizer code with syndrome s, maximum-likelihood decoding selects:

$$\ell^* = \arg\max_{\ell}\ p(\ell \,|\, s) = \arg\max_{\ell}\ p(s \,|\, \ell)\, p(\ell). \tag{49}$$

This is classical Bayes' rule—the classical limit (axiom (iv) of Theorem 1) of the Petz retrodiction functor. For the full quantum problem, the Petz map $\mathcal{R}_{\sigma_{\mathcal{C}},\mathcal{N}}$ generalizes ML by retrodicting the full quantum state including coherences.

The Retrodiction Gap Hierarchy

For decoder $\mathcal{D}$, define:

$$\delta_{\mathcal{D}} = D\Big(\tilde{\mathcal{R}}_{\sigma_{\mathcal{C}},\mathcal{N}} \,\big\|\, \mathcal{D}\Big), \tag{50}$$

measuring the distinguishability from optimal retrodiction. The established hierarchy [14] becomes:

$$0 = \delta_{\rm ML} < \delta_{\rm NN} < \delta_{\rm CM} < \delta_{\rm MWPM} < \delta_{\rm UF}. \tag{51}$$

The bound from the main text gives:

$$F(\rho,\, \mathcal{D}\circ\mathcal{N}(\rho))^2 \geq F(\rho,\, \tilde{\mathcal{R}}\circ\mathcal{N}(\rho))^2 \cdot e^{-\delta_{\mathcal{D}}}. \tag{52}$$

The Computational Entropy Conjecture

We conjecture $\delta_{\mathcal{D}} \leq c\,\sigma_{\mathcal{D}}$, where $\sigma_{\mathcal{D}}$ is the thermodynamic entropy produced during the decoder's computation. The motivation is Landauer's principle: erasing correlations during decoding costs $k_{\rm B}T\ln 2$ per bit and simultaneously degrades retrodictive information.

Heuristic support:

- Union-Find: greedy local rules discard long-range correlations (high σ, large δ).
- MWPM: global matching but ignores error correlations (moderate σ and δ).
- Neural networks (AlphaQubit [15]): learn correlations from data, potentially lower δ.
- ML/tensor-network: full probabilistic inference, minimal information erasure ($\delta \approx 0$).

Honest Assessment

Preliminary numerical experiments on $d = 3$ surface codes show that decoder *expressiveness* (the ability to represent complex correlations) dominates over thermodynamic efficiency in the small-code regime. A highly dissipative but expressive neural network outperforms a thermodynamically efficient but inexpressive decoder. We expect the conjecture to become relevant at $d \gtrsim 15$–20, where the syndrome space grows exponentially and efficient information compression becomes the bottleneck.

Even without the thermodynamic conjecture, the retrodiction-gap framework provides independent value: $\delta_{\mathcal{D}}$ is a decoder-independent metric computable from output statistics, offering a unified lens for comparing decoders of entirely different architectures.

S11. TEMPORAL ASYMMETRY AND THE EQUIVALENCE CHAIN

We prove the equivalence chain stated in the main text and develop the temporal asymmetry parameter τ.

Definition and Basic Properties

For a quantum channel $\mathcal{N}$, faithful reference state σ, and input ρ, define:

$$\tau(\rho, \sigma, \mathcal{N}) = 1 - F\big(\rho,\, \tilde{\mathcal{R}}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho))\big), \tag{53}$$

where $\tilde{\mathcal{R}}$ is the rotated Petz map of Junge et al. [4].

Basic properties:

1. $0 \leq \tau \leq 1$ (from $0 \leq F \leq 1$).

2. $\tau = 0$ if and only if exact Petz recovery: $\tilde{\mathcal{R}}_{\sigma,\mathcal{N}} \circ \mathcal{N}(\rho) = \rho$.

3. τ is continuous in all arguments (by continuity of fidelity).

4. τ is monotone under composition: for composable $\mathcal{N}_1, \mathcal{N}_2$, $\tau(\rho, \sigma, \mathcal{N}_2 \circ \mathcal{N}_1) \geq \tau(\rho, \sigma, \mathcal{N}_1)$, since additional processing cannot improve recovery.

The Equivalence Chain: Proof

We prove that for a faithful σ and channel $\mathcal{N}$ with Stinespring dilation $V : \mathcal{H}_A \to \mathcal{H}_B \otimes \mathcal{H}_E$, the following are equivalent:
(a) $\tau = 0$ (exact Petz recovery).
(b) $\Delta D = 0$ (DPI saturated).
(c) $I(A; E|B)_\omega = 0$ (quantum Markov chain).
(d) $\Sigma = 0$ (zero entropy production, for thermal σ).
(e) Detailed balance: $\mathcal{R}_{\sigma,\mathcal{N}} = \mathcal{N}$ (the channel is its own Petz reversal).

(a) ⇔ (b): This is the Petz sufficiency theorem [2]. Exact recovery $F = 1$ holds if and only if $D(\rho\|\sigma) = D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$ for all ρ in the support of σ, i.e., $\Delta D = 0$.

(b) ⇔ (c): By Fawzi–Renner [6], $I(A; E|B)_\omega = 0$ if and only if ω_{ABE} is a quantum Markov chain $A - B - E$. By the Stinespring structure, this is equivalent to perfect recoverability of A from B, i.e., $\Delta D = 0$.

(b) ⇔ (d) (for Gibbs σ): When $\sigma = e^{-\beta H}/Z$, the entropy production $\Sigma = \beta W_{\text{diss}} + D_{\text{mismatch}}$ by Kolchinsky–Wolpert [7]. $\Sigma = 0$ requires both $W_{\text{diss}} = 0$ and zero mismatch, which for the Gibbs reference means $\Delta D = \beta W_{\text{diss}} = 0$. The converse follows from $\Sigma \leq \Delta D$ (second law, Reeb–Wolf [8]).

(d) ⇔ (e): By Kwon–Kim [9]: the Crooks ratio $P_F/P_R = \exp(\Sigma/k_{\text{B}})$. Detailed balance ($\mathcal{R}_{\sigma,\mathcal{N}} = \mathcal{N}$, i.e., the channel equals its own reversal) holds if and only if $P_F = P_R$ for all trajectories, i.e., $\Sigma = 0$.

This completes the chain. □

Bound: $\tau \leq 1 - \exp(-\Sigma/2)$

From Theorem 3 of the main text (master inequality chain):

$$F^2 \geq \exp(-\Sigma). \tag{54}$$

Since $F \geq 0$ and $\exp(-x/2) \geq 0$, taking the square root:

$$F \geq \exp(-\Sigma/2). \tag{55}$$

Therefore:

$$\tau = 1 - F \leq 1 - \exp(-\Sigma/2). \tag{56}$$

For small Σ: Taylor-expanding gives $\tau \lesssim \Sigma/2$. For large Σ: $\tau \to 1$. The bound is tight when $F^2 = \exp(-\Sigma)$ (saturation of the Junge bound).

Connection to the Crooks Fluctuation Theorem

The Crooks theorem [17] states:

$$\frac{P_F(w)}{P_R(-w)} = \exp\big(\beta(w - \Delta F)\big). \tag{57}$$

For a cyclic process ($\Delta F = 0$), averaging over trajectories: $\langle P_F/P_R \rangle = \exp(\langle \beta W \rangle) = \exp(\Sigma/k_{\text{B}})$. When $\Sigma = 0$: $P_F = P_R$ pointwise. Forward and reverse processes are equally probable for every trajectory.

The temporal asymmetry parameter τ provides a *state-level* (rather than trajectory-level) measure of this asymmetry. While the Crooks ratio is defined over stochastic trajectories, τ is defined for a single input state ρ and measures whether that state can be recovered after the channel. The connection is: $\tau = 0$ implies $\Sigma = 0$ implies $P_F = P_R$ (the process is its own time reversal).

This is the precise mathematical content of the physical insight: in a zero-entropy-production system, there is no statistical distinction between the forward and reverse time directions.

The Retrodiction Landauer Principle

The energy cost of retrodiction—inferring the pre-channel state from the post-channel state—is bounded by the entropy produced during the forward process.

Theorem 2 (Retrodiction Landauer Principle)**.** *For a channel $\mathcal{N}$ with thermal reference state $\sigma_\beta = e^{-\beta H}/Z$, the minimum energy cost of any physical implementation of approximate retrodiction satisfies:*

$$E_{\text{retro}} \geq k_{\text{B}} T \cdot \Sigma. \tag{58}$$

Derivation.—By Landauer's principle [8], any computation that reduces uncertainty requires energy dissipation $\geq k_{\text{B}} T \ln 2$ per bit erased. Retrodiction from $\mathcal{N}(\rho)$ to ρ requires recovering at least ΔD nats of distinguishability. The energy cost of this recovery is:

$$E_{\text{retro}} \geq k_{\text{B}} T \cdot \Delta D. \tag{59}$$

By the master chain, $\Delta D \geq \Sigma$, giving Eq. (58).

Physical consequences:

- $\Sigma = 0$ (closed system): retrodiction is thermodynamically free. The past is as accessible as the future.

- $\Sigma > 0$ (open system): retrodiction has a minimum energy cost proportional to the entropy produced. The past becomes energetically expensive to access.

- The bound is tight when $\Delta D = \Sigma$ (all information loss is dissipative). □

The Quantum Eraser as the $\tau = 0$ Special Case

For a pure bipartite state $|\psi\rangle_{SE}$ with $\mathcal{N} = \mathrm{Tr}_E$:

- $\sigma = |\psi\rangle\langle\psi|_{SE}$ (pure).
- $D(\sigma\|\sigma) = 0$ and $D(\mathcal{N}(\sigma)\|\mathcal{N}(\sigma)) = 0$, so $\Delta D = 0$ trivially.
- By the equivalence chain: $\Delta D = 0 \Rightarrow I(A; E|B) = 0 \Rightarrow \Sigma = 0 \Rightarrow \tau = 0$.
- $F = 1$: exact recovery, as computed in Sec. S1.

The quantum eraser operates in the $\tau = 0$ regime. The idler measurement "works" not because of backward causation but because the signal-idler pair forms a closed system with zero entropy production. In such a system, the Crooks ratio is exactly 1, and there is no statistical distinction between measuring the signal first or the idler first.

The "need to wait" for the idler measurement arises only in the *open-system description*: when we trace out the idler and describe only the signal, we have $\tau > 0$ for the signal alone. The full bipartite system has $\tau = 0$, but the subsystem has $\tau > 0$. Recovering $\tau = 0$ requires accessing the idler—this is the operational meaning of "waiting for the future measurement."

S12. NUMERICAL VERIFICATION: DESIGN RATIONALE AND RESULTS

In this section we describe numerical experiments designed to test the *new predictions* of our framework (Predictions 1 and 2 of the main text), and explain why these experiments go beyond verifying known mathematics.

Why Not Simply Verify the DPI Bound?

The individual inequalities in the master chain (8) of the main text — $F^2 \geq \exp(-\Delta D)$ (Junge et al. [4]), $I(A; E|B) \geq -\log F^2$ (Fawzi–Renner [6]), and $\Delta D \geq \Sigma$ (Reeb–Wolf [8]) — are rigorously proven mathematical results. Computing them numerically for specific channels and confirming that they hold is analogous to verifying that $2 + 2 = 4$ on a calculator: it confirms implementation correctness but does not test any physical prediction.

Our numerical experiments are therefore designed *not* to recheck these known bounds, but to test consequences that follow specifically from the *uniqueness* of the Petz map as a retrodiction functor (Theorem 1) and from the *structure* of the master inequality chain when applied to error correction.

Experiment 1: Decoder Retrodiction Hierarchy

Known: Different QEC decoders (ML, MWPM, Union-Find, etc.) have different logical error rates [14]. This performance ordering is well-established empirically.

New prediction: The retrodiction gap $\delta_{\mathcal{D}} = D(\tilde{\mathcal{R}}_{\sigma,\mathcal{N}}\|\mathcal{D})$ should *quantitatively rank-order* decoder performance. Specifically, decoders with smaller $\delta_{\mathcal{D}}$ must have higher recovery fidelity.

Why this is not obvious: If the Petz map were merely *one of many* good recovery maps, then $\delta_{\mathcal{D}}$ (the divergence from the Petz map) would have no reason to predict decoder quality — a decoder could differ greatly from the Petz map yet still perform well by approximating some other good recovery map. The prediction that $\delta_{\mathcal{D}}$ rank-orders performance follows specifically from the *uniqueness* of the Petz map (Theorem 1): since it is the *only* Bayesian-consistent retrodiction functor, any sufficiently good decoder must converge toward it.

Numerical protocol:

1. Fix a surface code of distance d under depolarizing noise at rate p.
2. Implement four decoders: maximum-likelihood (ML, via exhaustive enumeration for small d), minimum-weight perfect matching (MWPM), Union-Find (UF), and a simple lookup-table decoder.
3. For each decoder $\mathcal{D}$, compute the effective channel $\mathcal{D} \circ \mathcal{N}$ by Monte Carlo sampling over error configurations.
4. Compute $\delta_{\mathcal{D}} = D(\tilde{\mathcal{R}}_{\sigma,\mathcal{N}}\|\mathcal{D})$ via the channel divergence between the Petz decoder and $\mathcal{D}$.
5. Compare the ordering $\delta_{\mathrm{ML}} < \delta_{\mathrm{MWPM}} < \delta_{\mathrm{UF}} < \delta_{\mathrm{lookup}}$ against the known performance ordering.

What would falsify the prediction: If a decoder with *larger* $\delta_{\mathcal{D}}$ consistently outperforms one with *smaller* $\delta_{\mathcal{D}}$, the retrodiction hierarchy interpretation would be falsified. This would indicate that the Petz map's uniqueness as a retrodiction functor does not constrain the structure of practical decoders.

Experiment 2: Post-Selection Thermodynamic Filtering

Known: Post-selection on syndrome outcomes can improve logical error rates in QEC [10, 13]. The improvement depends on the rejection fraction ε and the code distance d.

New prediction: The improvement factor obeys $\ln R(\varepsilon, d) = \alpha(p)\, d + \beta(\varepsilon)$, where:

- $\alpha(p)$ depends on the noise rate p but is *independent of the decoder* $\mathcal{D}$;

- $\beta(\varepsilon)$ depends on the rejection fraction but is *independent of the code distance* d.

Why α is decoder-independent: The data-processing gap $\Delta D(s)$ for syndrome s is determined by the noise channel $\mathcal{N}$ and the code state $\sigma_{\mathcal{C}}$ alone — it measures how much distinguishability is lost under the noise, which is a property of the noise and the code, not of the decoder. The per-distance scaling $\alpha(p)$ inherits this decoder-independence because it is set by the distribution of $\Delta D(s)$ values, which is a property of the channel.

Numerical protocol:

1. Fix surface codes of distances $d = 3, 5, 7$ under depolarizing noise at rates $p \in \{0.01, 0.03, 0.05\}$.

2. For each (d, p), sample 10^5–10^6 syndrome outcomes and compute $\Delta D(s)$ for each.

3. Apply multiple decoders (MWPM, UF, lookup) to each syndrome.

4. For each decoder and rejection fraction $\varepsilon \in [0, 0.3]$, compute the post-selected logical error rate $p_{\text{fail}}^{(\varepsilon)}$ and the improvement factor $R(\varepsilon, d)$.

5. Fit $\ln R(\varepsilon, d) = \alpha(p)\, d + \beta(\varepsilon)$ and verify:

   (a) The factorization holds (linear in d with d-independent intercept);

   (b) $\alpha(p)$ is the same across different decoders.

What would falsify the prediction: If α depends significantly on the decoder — for example, if MWPM and Union-Find give different slopes α at the same noise rate — this would indicate that the thermodynamic filtering interpretation of post-selection is incomplete.

Relationship to the Quantum Eraser Calculation

The explicit quantum eraser calculation (Sec. S1) serves a different purpose from the numerical experiments above. The eraser calculation demonstrates that the Petz map formalism correctly reproduces the known physics of quantum erasure ($\tau = 0$ for a closed bipartite system). This is a *consistency check* of the formalism, not a test of a new prediction. The numerical experiments in this section, by contrast, test consequences of the framework that are *not known a priori* and could in principle be falsified.

S13. CAVEATS AND LIMITATIONS

Operational vs. Ontological Interpretation

The equivalence $\Sigma = 0 \Leftrightarrow \tau = 0$ is a mathematical theorem about quantum channels and recovery maps. It does *not* imply:

1. That "the future already exists" in a closed system (block universe interpretation). The theorem is agnostic about ontology.

2. That retrocausation is physically real. The Petz map is a mathematical tool for Bayesian inference, not a physical process that sends information backward in time.

What the theorem *does* establish: in a closed system ($\Sigma = 0$), the statistical predictions are identical whether one conditions on past or future data. This is the precise content of the ABL rule [22] and the two-state vector formalism, now derived from the Petz recovery formalism rather than postulated.

"Zero Entropy" is Entropy Production, Not State Entropy

The quantity Σ in our framework is the *entropy production* of the system-environment interaction, not the von Neumann entropy of the system state. Key distinctions:

- A system can have high state entropy ($S(\rho)$ large) but zero entropy production ($\Sigma = 0$) if it evolves unitarily.

- Conversely, a pure state ($S(\rho) = 0$) interacting with an environment can have $\Sigma > 0$.

- The relevant quantity for temporal asymmetry is Σ (process irreversibility), not $S(\rho)$ (state uncertainty).

Measurement Back-Action

Any projective measurement on a quantum system produces entropy: the measurement process couples the system to a macroscopic apparatus (environment), creating $\Sigma > 0$. Therefore:

- Verifying $\tau = 0$ directly requires non-demolition or weak measurements.

- Projective measurement itself breaks the closed-system condition.

- The quantum eraser "resolves" this by deferring the measurement to the idler subsystem, preserving the closed-system condition on the signal-idler pair until the erasure measurement is performed.

Non-Markovian Subtleties

For non-Markovian processes (system-environment memory effects) [21]:

- Σ can be temporarily negative (information backflow from environment to system).
- This corresponds to transient τ decrease: the system partially "reverses" its temporal asymmetry.
- The equivalence chain holds for CP-divisible (Markovian) processes. For non-Markovian dynamics, the chain holds at the level of the total process map but not necessarily at intermediate times.

What Is Novel vs. What Is Known

We are transparent about the novelty content:
Known individually:

- Petz recovery for unitary channels gives $F = 1$ [2].
- Crooks theorem: $\Sigma = 0 \Leftrightarrow$ reversible [17].
- Fawzi–Renner: $I(A; E|B) = 0 \Leftrightarrow$ quantum Markov chain [6].
- Decoherence as origin of the arrow of time [18, 19].
- Landi–Paternostro: $\Sigma = 0 \Leftrightarrow$ detailed balance [20].

Novel in this work:

1. The *full equivalence chain* connecting all five conditions ($\tau = 0$, $\Delta D = 0$, $I(A; E|B) = 0$, $\Sigma = 0$, detailed balance) as a single unified result.
2. The parameter τ as an explicit, computable measure of temporal asymmetry via Petz recovery fidelity.
3. The bound $\tau \leq 1 - \exp(-\Sigma/2)$ connecting temporal asymmetry to entropy production.
4. The retrodiction Landauer principle $E_{\text{retro}} \geq k_{\text{B}} T \Sigma$.
5. The interpretation of the quantum eraser as the $\tau = 0$ special case, with "waiting" explained by the open-system description of a closed bipartite system.

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